# IEOR160HW4

March 11, 2026

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[106]:
```
import numpy as np
import cvxpy as cp
import matplotlib.pyplot as plt
```

## 2 Problem 1

$$\min f(x) = e^{x_1-1} + e^{-x_1+1} + e^{x_2-2} + e^{-x_2+2} + e^{-x_3+3} + (x_1 - 2x_2 + 5x_3)^4$$

$$\nabla f(x) = \begin{pmatrix} e^{x_1-1} - e^{-x_1+1} + 4(x_1 - 2x_2 + 5x_3)^3 \\ e^{x_2-2} - e^{-x_2+2} - 9(x_1 - 2x_2 + 5x_3)^3 \\ -e^{x_3+3} + 20(x_1 - 2x_2 + 5x_3)^3 \end{pmatrix}$$

[107]:
```
# Write a code in CVX to find the global minimum x*

x1 = cp.Variable()
x2 = cp.Variable()
x3 = cp.Variable()

objective = cp.Minimize(
    cp.exp(x1-1) + cp.exp(-x1+1) +
    cp.exp(x2-2) + cp.exp(-x2+2) +
    cp.exp(x3-3) + cp.exp(-x3+3) +
    cp.power((x1 - 2*x2 + 5*x3), 4)
)

problem = cp.Problem(objective)
problem.solve()

x1_cp = x1.value
x2_cp = x2.value
x3_cp = x3.value

print(f"CVXPY Solution: x1 = {x1_cp}, x2 = {x2_cp}, x3 = {x3_cp}")
```

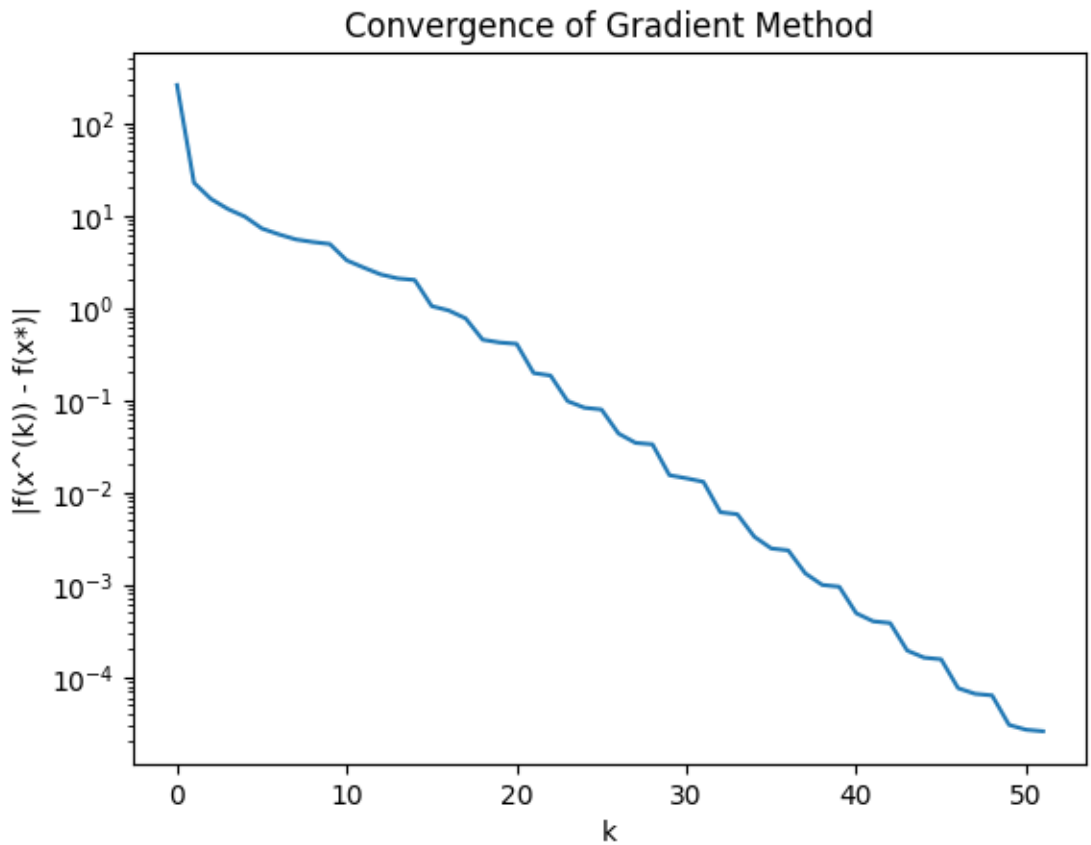
```
CVXPY Solution: x1 = 0.4573801355825401, x2 = 2.976508302346172, x3 =
1.2307154858979108
```

[108]:
```
# Then, write a code that implements the Gradient method with your choice of␣
↪the backtracking parameters.

x1 = 1
x2 = 1
x3 = 1

arr_g = []
arr_g.append([x1, x2, x3])


def f(x):
    x1, x2, x3 = x
    return (np.exp(x1-1) + np.exp(-x1+1) + np.exp(x2-2) + np.exp(-x2+2) + np.
↪exp(x3-3) + np.exp(-x3+3) + (x1 - 2*x2 + 5*x3)**4)


def calc_gradient(x):
    x1, x2, x3 = x

    dx1 = np.exp(x1-1) - np.exp(-x1+1) + 4 * (x1 - 2*x2 + 5*x3)**3 * 1
    dx2 = np.exp(x2-2) - np.exp(-x2+2) + 4 * (x1 - 2*x2 + 5*x3)**3 * (-2)
    dx3 = np.exp(x3-3) - np.exp(-x3+3) + 4 * (x1 - 2*x2 + 5*x3)**3 * (5)
    return np.array([dx1, dx2, dx3])


def step_size():
    # backtracking line search
    alpha = 0.05  # Step size reduction factor
    beta = 0.1  # Inital t

    while f(arr_g[-1] - beta * calc_gradient(arr_g[-1])) >= f(arr_g[-1]):
        beta *= alpha

    return beta


for i in range(51):
    gradient = calc_gradient(arr_g[-1])
    t = step_size()
    x_new = arr_g[-1] - t * gradient
    arr_g.append(x_new)

xstar = arr_g[-1]
```

```
print(f"Gradient Descent Solution: x1 = {xstar[0]}, x2 = {xstar[1]}, x3 =␣
↪{xstar[2]}")
```

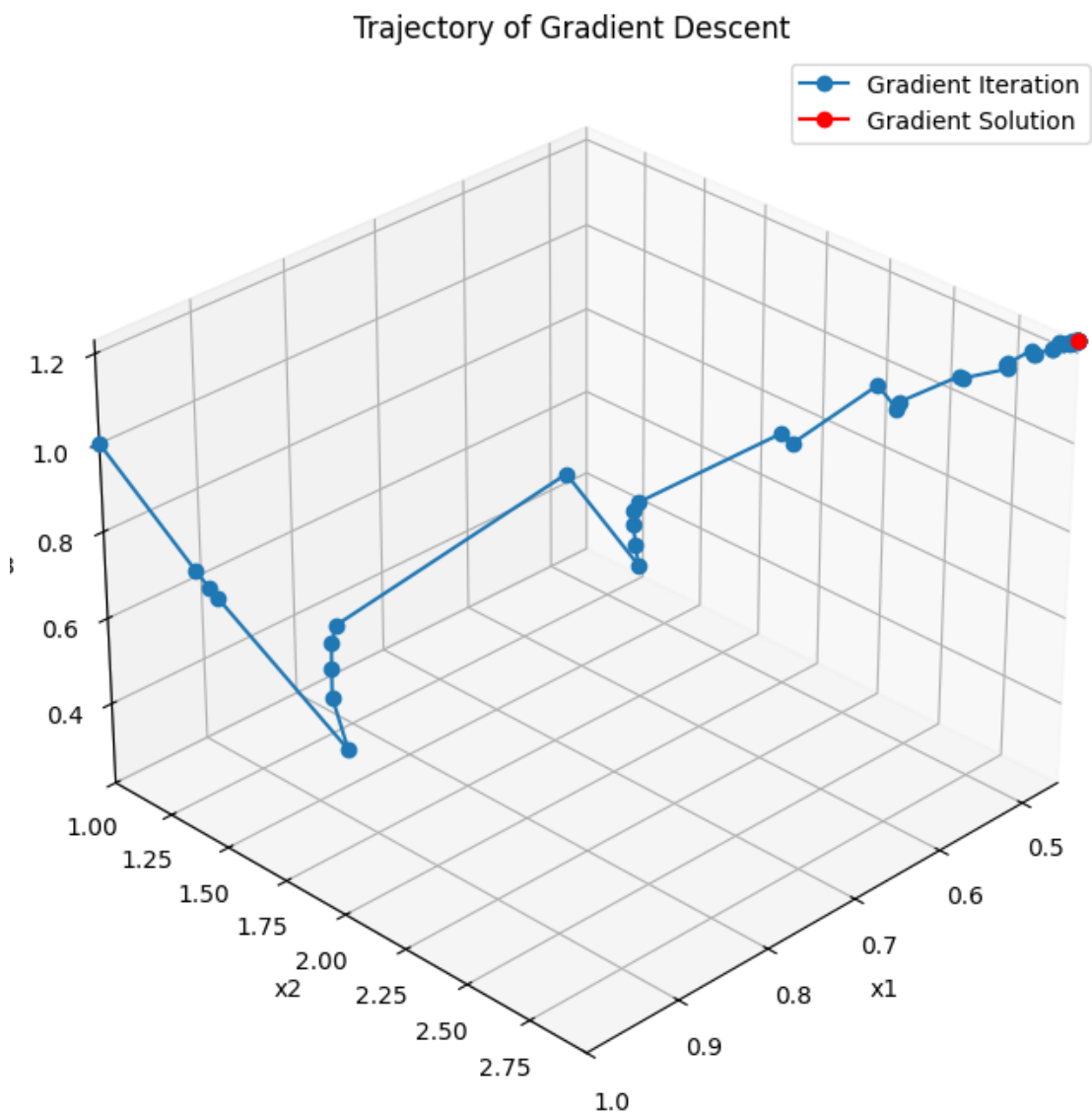
```
Gradient Descent Solution: x1 = 0.4566138962977194, x2 = 2.972826937170818, x3 =
1.2294275351541832
```

[109]:
```
# Draw |f(x(k)) - f(x*)| verses k for k = 0, 1, 2, . . . , 50 on alog-linear␣
↪plot.
# Show the trajectory of the points x(0), x(1), ..., x(50) in the 3-dimensional␣
↪(x1, x2, x3) plane.

f_vals = np.array([f(x) for x in arr_g])

x_star = np.array([x1_cp, x2_cp, x3_cp])
f_star = f(x_star)

errors = np.abs(f_vals - f_star)
k = np.arange(len(arr_g))

plt.semilogy(k, errors)

plt.xlabel("k")
plt.ylabel("|f(x^(k)) - f(x*)|")
plt.title("Convergence of Gradient Method")

plt.show()
```




[110]:
```
fig = plt.figure(figsize=(8, 8))

ax = fig.add_subplot(111, projection="3d")

arr_g = np.array(arr_g)

ax.plot(arr_g[:,0], arr_g[:,1], arr_g[:,2], marker="o", label = "Gradient␣
↪Iteration")

ax.plot(*xstar, marker="o", color="red", label = "Gradient Solution")

ax.set_xlabel("x1")
ax.set_ylabel("x2")
ax.set_zlabel("x3")
ax.set_title("Trajectory of Gradient Descent")

ax.set_xlim(min(arr_g[:,0]), max(arr_g[:,0]) )
ax.set_ylim(min(arr_g[:,1]), max(arr_g[:,1]) )
ax.set_zlim(min(arr_g[:,2]), max(arr_g[:,2]) )
```

```
ax.legend()

ax.view_init(azim=45)
plt.show()
```




The plot shows that the gradient descent algorithm does converge towards the actual solution, as we get to 4 digits of accuracy from its proposed solution. However, even after 50 iterations, the error plot suggests that the solution can be improved as the line has not flattened.

The trajectory graph showed that the gradient descent algorithm varied a lot in step length, with many small clusters.

## 3 Problem 2

Redo the previous problem with Newton's method.

$$\min f(x) = e^{x_1-1} + e^{-x_1+1} + e^{x_2-2} + e^{-x_2+2} + e^{-x_3+3} + (x_1 - 2x_2 + 5x_3)^4$$

$$\nabla f(x) = \begin{pmatrix} e^{x_1-1} - e^{-x_1+1} + 4(x_1 - 2x_2 + 5x_3)^3 \\ e^{x_2-2} - e^{-x_2+2} - 8(x_1 - 2x_2 + 5x_3)^3 \\ -e^{x_3+3} + 20(x_1 - 2x_2 + 5x_3)^3 \end{pmatrix}$$

$$\nabla^2 f(x) = \begin{pmatrix} e^{x_1-1} + e^{-x_1+1} + 12(x_1 - 2x_2 + 5x_3)^2 & -24(x_1 - 2x_2 + 5x_3)^2 & 60(x_1 - 2x_2 + 5x_3 \\ -24(x_1 - 2x_2 + 5x_3)^2 & e^{x_2-2} + e^{-x_2+2} + 48(x_1 - 2x_2 + 5x_3)^2 & -120(x_1 - 2x_2 + 5 \\ 60(x_1 - 2x_2 + 5x_3)^2 & -120(x_1 - 2x_2 + 5x_3)^2 & e^{x_3+3} + 300(x_1 - 2x_2 - \end{pmatrix}$$

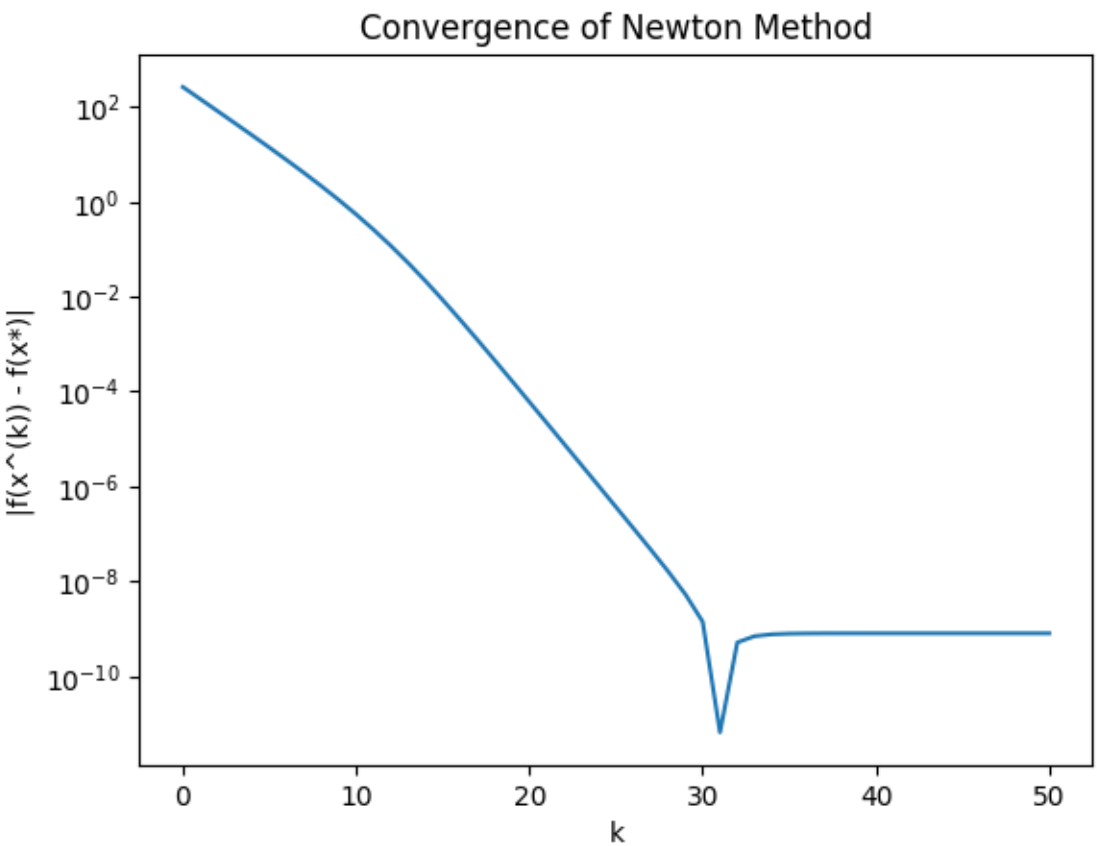
[111]:
```
x_0 = [1,1,1]
arr_n = []
arr_n.append(x_0)

# compute gradient (implemented in 1st part)

# compute hessian
def calc_hessian(x):

    x1, x2, x3 = x

    dx1x1 = np.exp(x1 - 1) + np.exp(-x1 +1) + 12 * (x1 - 2*x2 + 5*x3)**2
    dx1x2 = -24 * (x1 - 2*x2 + 5*x3)**2
    dx1x3 = 60 * (x1 - 2*x2 + 5*x3)**2

    dx2x1 = dx1x2
    dx2x2 = np.exp(x2 - 2) + np.exp(-x2 + 2) + 48 * (x1 - 2*x2 + 5*x3)**2
    dx2x3 = -120 * (x1 - 2*x2 + 5*x3)**2

    dx3x1 = dx1x3
    dx3x2 = dx2x3
    dx3x3 = np.exp(x3 - 3) + 300 * (x1 - 2*x2 + 5*x3)**2

    hessian = np.array([
        [dx1x1, dx1x2, dx1x3],
        [dx2x1, dx2x2, dx2x3],
        [dx3x1, dx3x2, dx3x3]
    ])

    return hessian
```

```python
# compute direction
def calc_direction(x):
    # delta = - H^-1 * g
    delta = np.linalg.solve(calc_hessian(x), -calc_gradient(x))
    return delta

# step size
def step_size(x):
    # backtracking
    alpha = 0.5
    beta = 0.4

    direction = calc_direction(x)
    tol = 1e-20
    while f(x + beta * direction) >= f(x) and beta > tol:
        beta *= alpha

    return beta

for i in range(50):
    # compute step size
    curr_x = arr_n[-1]

    new_x = curr_x + step_size(curr_x) * calc_direction(curr_x)
    arr_n.append(new_x)

x_star = arr_n[-1]
print(f"Newton's Method Solution: x1 = {x_star[0]}, x2 = {x_star[1]}, x3 =␣
↪{x_star[2]}")
```

```
Newton's Method Solution: x1 = 0.4573950392090808, x2 = 2.97649383192283, x3 =
1.2307065747966477
```

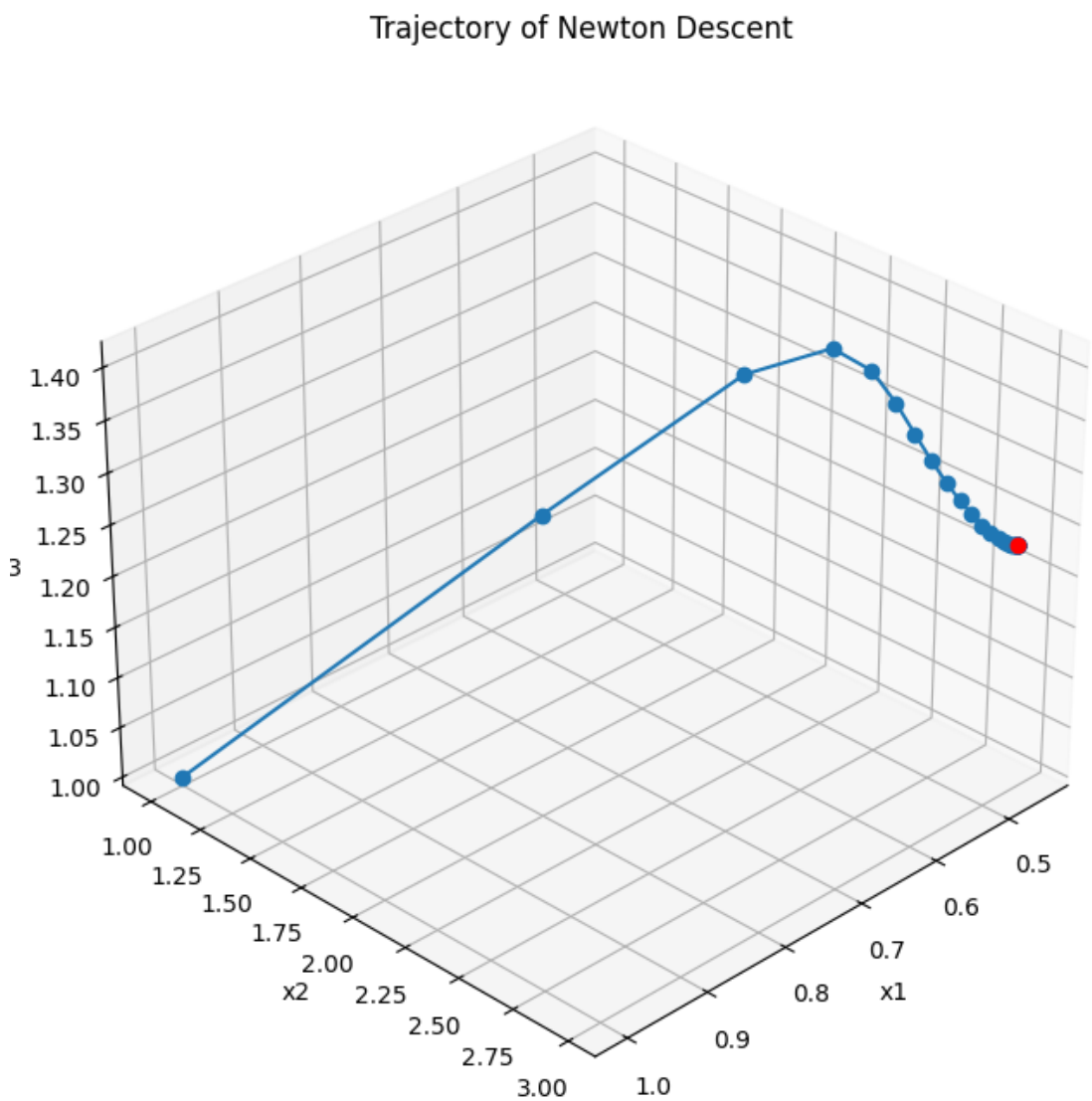
[112]:
```python
f_vals = np.array([f(x) for x in arr_n])

x_star = np.array([x1_cp, x2_cp, x3_cp])
f_star = f(x_star)

errors = np.abs(f_vals - f_star)
k = np.arange(len(arr_n))

plt.semilogy(k, errors)

plt.xlabel("k")
plt.ylabel("|f(x^(k)) - f(x*)|")
plt.title("Convergence of Newton Method")
```

```
plt.show()
```




[113]:
```
fig = plt.figure(figsize=(8, 8))
ax = fig.add_subplot(111, projection="3d")

arr_n = np.array(arr_n)

ax.plot(arr_n[:,0], arr_n[:,1], arr_n[:,2], marker="o")
ax.plot(*x_star, marker="o", color="red", label = "CVXPY Solution")

ax.set_xlabel("x1")
ax.set_ylabel("x2")
ax.set_zlabel("x3")
ax.set_title("Trajectory of Newton Descent")

ax.view_init(azim=45)
```

```
plt.show()
```




Newton's algorithm convergerd to the cvx solution extremely fast, at around 31 iterations. I had to add a tolerance term to the backtracking loop to stop an infinite loop as beta approached 0 without a better solution, which resulted in the dip. Afterwards, it could not improve the solution any further.

Newton's trajectory graph showed that the step size started large and steadily decreased for each iteration until convergence.

# 4 Problem 3:

$$\min_{x \in \mathbb{R}^n} -\sum_{i=1}^{n} log(1 - x_i^2) - \sum_{i=1}^{n} log(4 + a_i^T x)$$

[114]:
```
# CVX Solution:

np.random.seed(seed=42)

n = 2000

# generates 2000 random vectors of dim 2000
A = np.random.rand(n, n)

# 2000 variables
x = cp.Variable(n)

# obj func
func = cp.Minimize(-cp.sum(cp.log(1- x**2)) - cp.sum(cp.log(4 + A @ x)) )

problem = cp.Problem(func)

result = problem.solve()
```

[115]:
```
# 3 CVX
cvx_x_star = x.value

if problem.status == cp.OPTIMAL:
    # 1. CVXPY Status
    print(f"Status: {problem.status}")

    # 2. Optimal Value f(x*)
    # Note: problem.value reflects the Maximization.
    # If your original problem was a minimization, use -problem.value
    print(f"Optimal Value f(x*): {problem.value}")

    # 3. Norm of x*
    x_norm = np.linalg.norm(x.value)
    print(f"Norm of x* (||x||_2): {x_norm}")

else:
    print(f"Solver was not successful. Status: {problem.status}")
```

```
Status: optimal
Optimal Value f(x*): -11919.749025718062
Norm of x* (||x||_2): 25.75919281439763
```

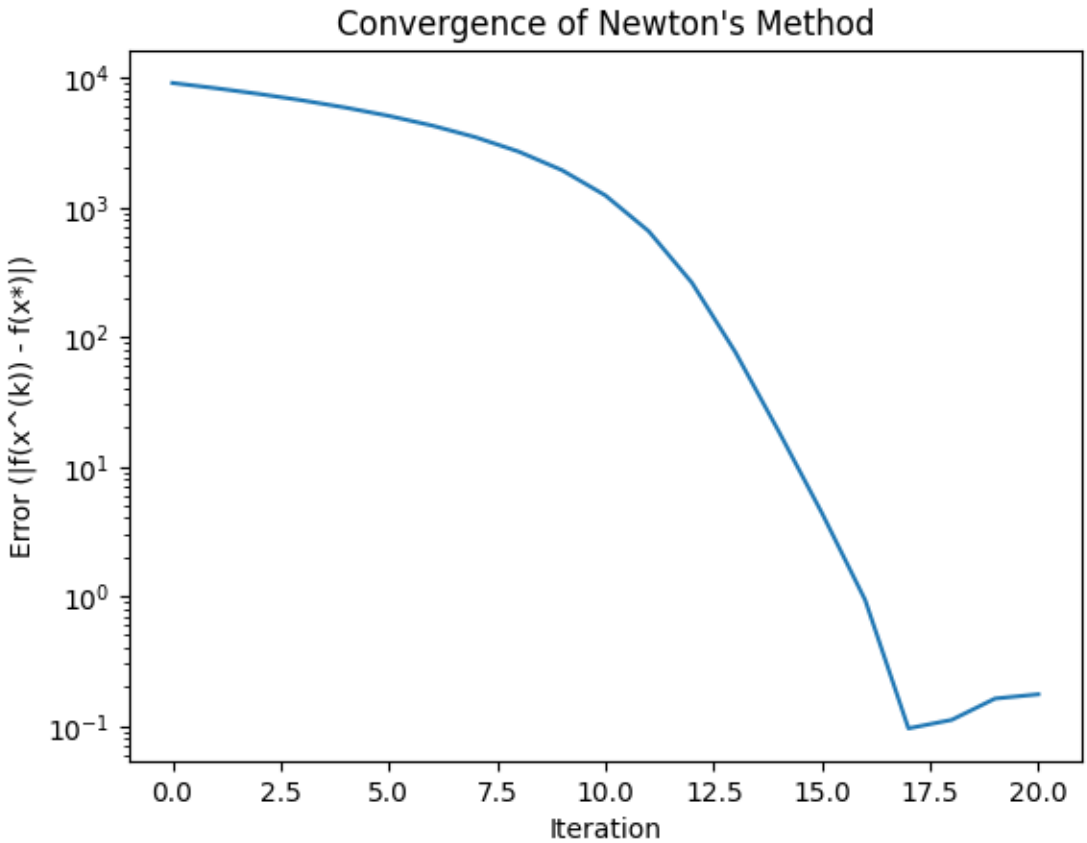
[ ]:
```
# 3 Newton's

np.random.seed(0)

n = 2000

A = np.random.rand(n, n)

def f(x):
    term1 = -np.sum(np.log(1 - x**2))
    term2 = -np.sum(np.log(4 + A @ x))
    return term1 + term2

def grad(x) -> np.ndarray:
    #calculate the gradient of the function

    # chain rule for a generic x_i
    grad_term1 = 2*x / (1-x**2)

    # term 2 is - sum (ai / (4 + ai @ x))
    # in general it is - sum (A[i] / (4 + A[i] @ x)) for i in range(n), or -A.T␣
↪@ (1/ (4 + A @ x))
    grad_term2 = -A.T @ (1/ (4 + A @ x))

    return grad_term1 + grad_term2

def hess(x) -> np.ndarray:
    # calculate the hessian of the function

    # Term 1
    # since the derivative for x_1 of term 1 is 2*x_1 / (1-x_1**2), it only␣
↪depends on 1 x
    # Therefore hessian is a diagonal matrix of the second derivs
    diag = (2 * (1 + x**2)) / (1 - x**2)**2
    hessian_1 = np.diag(diag)

    # Term 2
    # Hessian of term 2 is sum(ai  aiT / (4 + a_i x)^2) for each i
    # This is same as A.T @ diag(1/(4 + A @ x)**2) @ A because we need to␣
↪multiply each 1/(4+ai x)^2 by aiaiT
    d = 1/(4 + A @ x)**2
    hessian_2 = A.T @ np.diag(d) @ A

    return hessian_1 + hessian_2


# compute direction
```

```python
def calc_direction(x):
    # delta = - H^-1 * g
    delta = np.linalg.solve(hess(x), -grad(x))
    return delta

# step size
def step_size(x):
    # backtracking
    alpha = 0.1
    beta = 0.5

    direction = calc_direction(x)
    tol = 1e-50
    while f(x + beta * direction) >= f(x) and beta > tol:
        beta *= alpha

    delta = beta * direction
    return delta, beta

x_k = []
x0 = np.zeros(n)
x_k.append(x0)

for i in range(101):
    # compute step size
    curr_x = x_k[-1]
    delta, beta = step_size(curr_x)
    new_x = curr_x + delta
    x_k.append(new_x)

x_star = x_k[-1]
print(f"Newton's Method Solution: {f(x_star)}")
```

```
Newton's Method Solution: -11919.68361822699
```

[120]:
```python
f_vals = np.array([f(x) for x in x_k])

# f_star = problem.value
f_star = f(cvx_x_star)
errors = np.abs(f_vals - f_star)

k = np.arange(len(x_k))

plt.semilogy(k, errors)
plt.xlabel('Iteration')
plt.ylabel('Error (|f(x^(k)) - f(x*)|)')
plt.title('Convergence of Newton\'s Method')
```

```
plt.show()
```




Here, Newton's algorithm seems to converge is about 16 iterations, which took my computer around 11.6s to run. Comparatively, CVX took around 40 seconds to run. This indicates that our algorithm is much faster in computing the optimal solution. Again, there is a dip in the graph because newton's backtracking made beta go to 0, which I had to limit to prevent an infinite loop.